

Breaking Points: How Momentum Shifts the Scoreboard in Tennis

Summary

The Wimbledon Tennis Championships, a pinnacle of global tennis prestige, bore witness to a historic moment in 2023. Carlos Alcaraz, a 20-year-old phenom, emerged victorious against Novak Djokovic, a titan with seven Wimbledon victories. This match showcased an intangible momentum that seemed immeasurable, highlighting the importance of strategic momentum management for players and coaches alike. To decode player momentum and identify the pivotal factors influencing match outcomes, we constructed a quantitative model based on the 2023 tournament's data.

For Task 1, we employed a **Performance Judgment Model based on Logistic Regression** and a **Performance Scoring Model based on the Entropy-TOPSIS**. Specifically, by hypothesizing that the scoring side demonstrates superior performance, we analyzed the indicators affecting the scorer's performance. Through logistic regression, we determined the functional relationship between these indicators and the performance outcome, facilitating an assessment of player performance. Subsequently, the **TOPSIS entropy weighting evaluation model** was employed to score players' performances. Given redundant indicators that represent the model's reliability, we adjusted the final scores by incorporating these indicators as bonus or penalty points based on the TOPSIS scores. This nuanced approach allowed us to accurately gauge player performance, achieving a model **accuracy rate of 73%** in performance judgments and **effectively measuring specific player performances**.

For Task 2, we employed a **Momentum Prediction Model based on Long Short-Term Memory (LSTM)**. Recognizing the significant impact of time series on player momentum, we utilized LSTM to learn the influence of past performances on current scores, augmented with time-correlated factors for auxiliary training, aiming to forecast future performance. The model's predictions were then smoothed to deduce player momentum. After 2000 iterations, our model achieved a root mean square error (RMSE) of **only 8.134**, aligning closely with the actual match progressions.

For Task 3, we employed a **Factor Ranking Model based on Random Forest and Spearman's Rank Correlation Analysis**. Specifically, we assume that the intersection points of the two players' momentum curves signify crucial **turning points** in the match's flow. Through **Random Forest**, we predicted the intersection points of the momentum curves, obtaining a preliminary ranking of influencing factors. Subsequently, Spearman's rank **correlation analysis** was employed to further refine and validate the ranking of these factors, confirming the reliability of our rankings. We ultimately found that the **server's advantage** and the **distance covered by the players** have the most significant impact on the match's progression.

For Task 4, we conducted **validation** and **generalizability analysis** for our model. Initially, we applied our model to other men's and women's tennis matches, confirming its reliability and accuracy. **These results indicates a better generalizability of our model**. Finally, we explored and compared the differing factors influencing match fluctuations between men's and women's matches.

Our study highlights momentum's role in tennis, providing strategies for coaches and players to navigate match dynamics effectively.

Keywords: Logistic Regression; TOPSIS; LSTM; Random Forest; Correlation Analysis.

Contents

1	Introduction	3
1.1	Problem Background	3
1.2	Restatement of the Problem	3
1.3	Our Work	3
2	Assumptions and Notations	4
2.1	Assumptions	4
2.2	Notations	4
3	Data Preprocessing	5
4	Task 1: Performance Evaluation	5
4.1	Model1: Performance Judgment based on Logistic Regression	5
4.1.1	Logistic Regression	6
4.1.2	Indicator Establishment	6
4.2	Results and Analysis for Model1	7
4.3	Model2: Performance Scoring Based on Entropy-TOPSIS	8
4.3.1	Entropy Weight Method	8
4.3.2	Technique for Order of Preference by Similarity to Ideal Solution	9
4.3.3	Indicator Establishment	10
4.3.4	Bonus & Penalty Mechanism	10
4.4	Results and Analysis for Model2	11
5	Task2: Momentum Evaluation	12
5.1	Momentum Prediction Model Based on LSTM	12
5.1.1	Long-Short Term Memory	12
5.1.2	Indicator Establishment	13
5.2	Results and Analysis	14
6	Task3: Momentum Swing Prediction	16
6.1	Random Forest	16
6.2	Spearman's Rank Correlation Analysis	17
6.3	Indicator Establishment	17
6.4	Results and Analysis	18
7	Task 4: Validation and Generalization Analysis	20
7.1	Model Validation on Other Matches	20
7.2	Generalization Analysis on Women's Matches	21
7.3	Model Improving	22
8	MEMO	23
	References	24

1 Introduction

1.1 Problem Background

Wimbledon Championships is one of the most high-profile tennis tournament in the world. The 2023 Wimbledon Championships' final was a historical match, that the 20-year-old new star Carlos Alcaraz defeated the 7-time Wimbledon champion Novak Djokovic to win his first Wimbledon men's singles title, breaking Djokovic's 34-match Wimbledon winning streak. During the match, there was a clear shift in dominance. Djokovic initially dominated, winning the first set 6-1. The second set was tightly contested, with Carlos Alcaraz winning 7-6. Alcaraz then took control, winning the next set 6-1. However, Djokovic regained the upper hand in the following set with a 6-3 win. In the final set, momentum shifted back to Alcaraz, who clinched the match with a 6-4 victory. The incredible swings, sometimes for many points or even games, that occurred in the player who seemed to have the advantage are often attributed to "momentum."

1.2 Restatement of the Problem

Given the background information and the constraints outlined in the problem statement, we aim to analyze the data from every point in all Wimbledon 2023 Men's matches to address the following objectives:

- Develop a model to measure player performance at specific times, determining precisely how much better one player performs over another, and utilize this model to visualize the match's flow through detailed analysis.
- Create a model to measure players' momentum and assess its impact on the outcome of matches. Investigate whether shifts in play and streaks of success are random or patterned.
- Construct a prediction model to forecast future turning points in a match. Use a correlation model to gauge the effect of specific indicators on match trends, identify the most critical indicators influencing trend shifts, and offer recommendations for players to maintain their advantage during a game.
- Conduct analyses using the pre-proposed model on additional matches to evaluate its generalizability. Suggest further indicators that could enhance the model's accuracy.

1.3 Our Work

We have proposed four models to uncover the latent momentum within players. The structure of our paper is illustrated in the Figure 1. The remainder of this paper is organized as follows. In Section 2, we introduce the underlying assumptions and rationale, specifying the common variables in our formulas. In Section 3, we delve into data preprocessing before modeling. In Section 4, we evaluate player performance. Section 5 entails a quantitative analysis of momentum, while Section 6 focuses on predicting changes in match advantage. In Section 7, we subject our models to reliability checks and generalization analysis. Finally, we summarize the results of the entire paper.

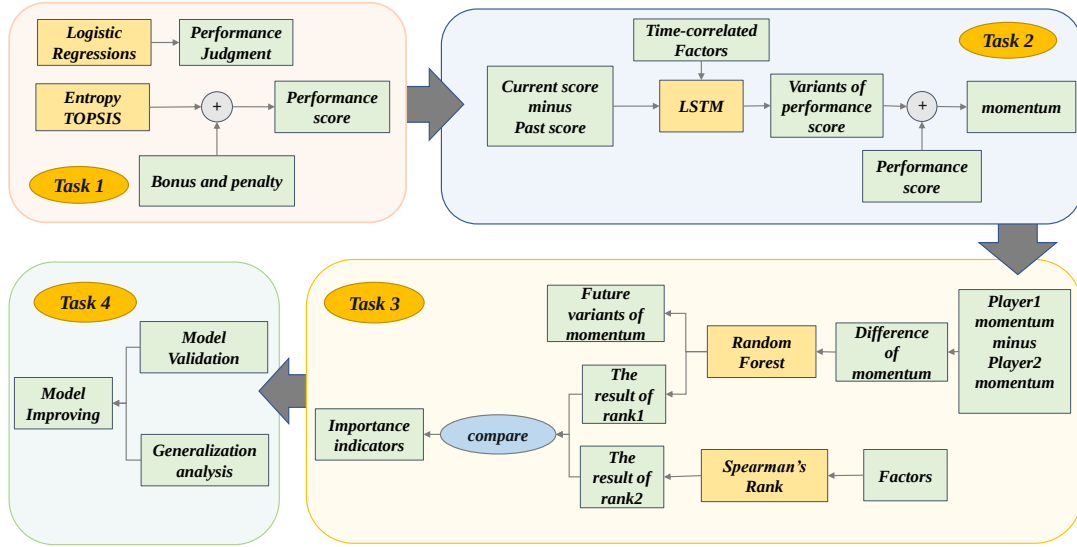


Figure 1: Visualization our work.

2 Assumptions and Notations

2.1 Assumptions

Considering that practical problems always contain many complex indicators, first of all, we need to make reasonable assumptions to simplify the model. The assumptions are summarized as follows:

- Assumption 1:** Assume that the performance of the player is unaffected by weather conditions, such as temperature, humidity, wind, or precipitation.
- Assumption 2:** Assuming that a player's performance is unaffected by the court surface requires data across different surface types to support this assumption.
- Assumption 3:** Assume that the player's physical state remains stable during matches, meaning no significant changes due to injuries, fatigue, or other health issues affecting their performance.
- Assumption 4:** Assume that the player's training methods and match strategies remain unchanged during the observation period, or that these changes can be quantified with available data.

Additional assumptions are made to simplify analysis for individual sections. These assumptions will be discussed at the appropriate locations.

2.2 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper.

Symbol	Description
x_a	Indicators for the logistic regression model.
x_b	Quantized indicators for TOPSIS model.
x_c	Indicators for the Random Forest model.
F_{base}	Base score of performance.
F_{bonus}	Bonus score of performance.
F_{score}	Evaluated performance score.
M	Evaluated momentum.
M_{diff}	Difference in momentum between two players.

3 Data Preprocessing

The given data of Wimbledon 2023 Gentlemen’s singles matches contained several error data that requires preprocessing. We addressed these inaccuracies as follows:

(i) *Lost data of serve speed and rally count in match '2023-wimbledon-1310'*: We fix this lost data by calculating their averages from other matches and substituting these averages into the missing values.

(ii) *Different Representations of Fraction in tie-breakers*: The given data represented scores using 15, 30, 40, and AD, but used a 1-7 format for tie-breakers. Such differential representations can mislead the model. We standardized these scores by converting tie-breaker scores to a range of 10-70 .

(iii) *Quantifying Non-Numeric Metrics*: Several non-numeric data like 'AD' in 'p1/p2_score', 'NA' in general 'speed_mph', 'winner_shot_type', 'serve_width', 'serve_depth' and 'return_depth'. We quantified these data by assigning values of 0, 1, and -1 based on the context. Specifically, for 'AD' in scores, we assigned a value of 55, indicating the next level beyond a score of 40.

4 Task 1: Performance Evaluation

To solve the problem of which player performs in a given time and quantifying their superiority. We first propose **Model1**, which utilizes the logistic regression method to predict Player 1’s chance of winning, thereby assessing **who performs better**. Subsequently, **Model2**, based on the Entropy-TOPSIS method and augmented with a bonus and penalty mechanism, is proposed to precisely quantify **the extent of their superior performance**.

4.1 Model1: Performance Judgment based on Logistic Regression

In this model, we evaluate who performances better by predicting the winning chance of Player 1 against Player 2. The assumption is that a higher winning chance indicates better performance. To this end, a **Logistic Regression** approach is employed to forecast the winning chance using **five distinct indicators** from both competitors.

4.1.1 Logistic Regression

Logistic regression[1][2] is a pivotal statistical method used for binary classification problems, mapping input variables to predict a binary outcome. Its application spans various fields such as medicine, social sciences, and machine learning.

The heart of logistic regression is the logistic function, modeling the probability that a dependent variable is one of two possible outcomes. It is expressed as:

$$P_{win_player1}(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}, \quad (1)$$

where $P_{win_player1}(Y = 1)$ denotes the probability of the dependent variable Y being 1, e is the base of the natural logarithm, $X_1, X_2, \dots, X_n$ represent the independent variables, and $\beta_0, \beta_1, \dots, \beta_n$ are the coefficients to be estimated by the model.

The logistic regression model estimates its coefficients using maximum likelihood estimation (MLE), aiming to maximize the likelihood of observing the sample data. The loss function used in logistic regression, also known as log loss, is given by:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(\sigma(\theta^T X^{(i)})) + (1 - y^{(i)}) \log(1 - \sigma(\theta^T X^{(i)}))], \quad (2)$$

here, $y^{(i)}$ represents the actual outcome of the i -th training example, and σ represents the logistic (sigmoid) function. θ is the parameters (coefficients) of the model.

This function measures the performance of the model, with $y^{(i)}$ being the actual outcome and $X^{(i)}$ being the features of the i -th training example. To minimize this function, the parameters θ are updated iteratively using gradient descent:

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}, \quad (3)$$

where α is the learning rate, and the gradient of $J(\theta)$ with respect to θ_j is:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m (\sigma(\theta^T X^{(i)}) - y^{(i)}) X_j^{(i)}, \quad (4)$$

where m represents the number of training examples. The iterative update process continues until the change in the loss function $J(\theta)$ between iterations is below a predetermined threshold, indicating convergence.

4.1.2 Indicator Establishment

To evaluate which player performs better in a specific time, we proposed a model $P_{win_player1}(x)$ to evaluate the player 1's winning probability. Here, we summarized five indicators used in this model, which are considered as the most important indicators for winning chance, showed as follows:

- x_{a1} : Indicates if Player 1 is serving.
- x_{a2} : Specifies whether it is the first or second serve.
- x_{a3} : The run distance of Player 1 during the point in meters.
- x_{a4} : The run distance of Player 2 during the point in meters.
- x_{a5} : The speed of the serve in miles per hour (mph).

4.2 Results and Analysis for Model1

We trained the model with a training set consisting of 30 matches' data, and validation test set consisting of data from 1 match. Then, we used the logistic regression maximum likelihood ratio calculation method to solve for the regression coefficients. By substituting the obtained regression coefficients into the established equation, we can obtain the following results.

$$P_{win_player1}(Y = 1) = \frac{1}{1 + e^{-(-0.078 + 1.443x_{a1} + 0.122x_{a2} - 0.243x_{a3} + 0.078x_{a4} + 0.238x_{a5})}} \quad (5)$$

We also listed these weights in Table 2 and visualized them in Figure 2, where the server plays the most important role in winning, while serving speed also has a positive effect on winning. Moreover, the running distance of Player 1 leads to a negative effect on winning, which can occur due to the physical exhaustion.

Table 2: Model weights and indicators

Weights	Indicator Name	Value
β_0	server	1.443
β_1	serve_no	0.122
β_2	p1_distance_run	-0.243
β_3	p2_distance_run	0.078
β_4	speed_mph	0.238
b	bias	-0.078

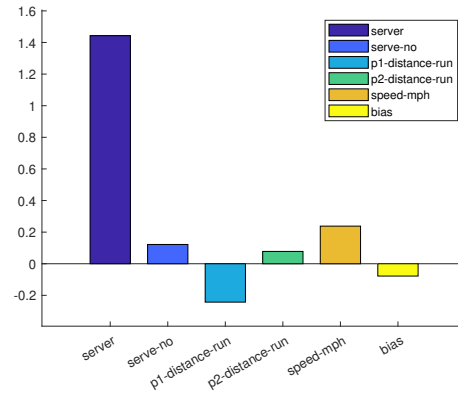


Figure 2: Bar chart of weights

We then evaluate **who performs better** by simply comparing the rounded results of the solved function $P_{win_player1}(x)$ to 0 or 1. Specifically, if the rounded result is 1 it shows that Player 1 has a higher winning chance, indicating Player 1 has a better performance. The final results are shown in Table4 where our model ultimately achieved a 73.0% of Accuracy, Recall and F1 Score. This shows that, our model can roughly predict the win chance of Player 1.

Table 3: Model Results.

Accuracy	Recall	Precision	F1 Score	AUC
73.0%	73.0%	73.0%	72.9%	79.0%

We further evaluate it with receiver operating characteristic (ROC) curve, a graphical plot used to show the diagnostic ability of binary classifiers. The ROC curve plots the true positive rate (TPR) on the X-axis against the false positive rate (FPR) on the Y-axis at various threshold settings, which are defined as follows:

$$TPR = \frac{TP}{TP + FN} \quad FPR = \frac{FP}{FP + TN}, \quad (6)$$

where: TP is the number of positive instances correctly classified. FN is the number of positive instances incorrectly classified as negative. FP is the number of negative instances incorrectly classified as positive. TN is the number of negative instances correctly classified.

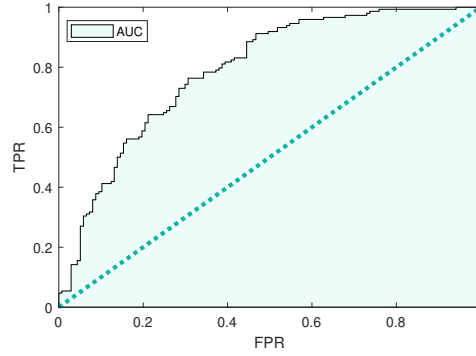


Figure 3: ROC curve.

As shown in Figure 3 most of the ROC curve is above the center line and achieved a 68.0% of the Area Under the Curve (AUC), indicating a good performance of a binary classifier of winning.

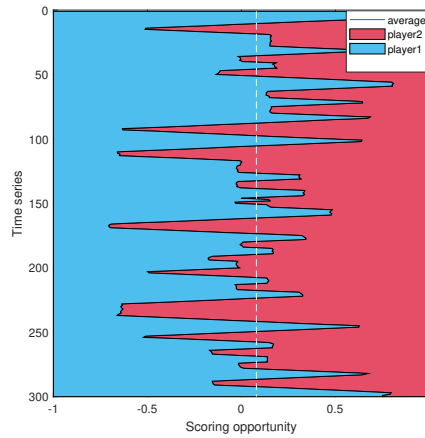


Figure 4: Visualization of winning chance.

We then visualize the predicted winning chance scores of Player 1 in the test set in Figure 4 as the game flow, where the blue area represents the winning chance of Player 1, while the red area representing Player 2. As the y axis indicating the time series, the results show that both two players have different winning chance in different time. Between the alternations of having better winning chances, it shows an obvious winning trend caused by the 'momentum'. Also, the average win chance of Player 1, the white line in the figure, is positive, which matches the ultimate win of Player 1 in this match.

4.3 Model2: Performance Scoring Based on Entropy-TOPSIS

We further propose Model2 to evaluate the actual specific performance, regardless of the initial advantages or disadvantages from serving. This model consist of two parts: (i) It first applies Entropy-TOPSIS method to solve a base performance score function $F_{base_score}(x_1)$ with preprocessed indicators regardless of the initial advantages. (ii) Then, it adds a bonus score through a bonus function $F_{bonus_score}(x_2)$ considering all conditions of winning or losing. We then compare the calculated performance score in a given time to evaluate **how much better they are performing**.

4.3.1 Entropy Weight Method

The Entropy method[3] serves as a foundation for quantifying the significance of each criterion within a decision matrix, thereby determining the weights in an objective manner. Firstly,

let there be m alternatives from dataset and n criteria, the decision matrix D is an $m \times n$ matrix where d_{ij} represents the performance of alternative i in terms of criterion j . Then, the decision matrix is normalized to allow comparisons across criteria using:

$$r_{ij} = \begin{cases} \frac{d_{ij}}{\sum_{i=1}^m d_{ij}}, & \text{for benefit criteria} \\ \frac{1/d_{ij}}{\sum_{i=1}^m 1/d_{ij}}, & \text{for cost criteria} \end{cases} \quad (7)$$

Then, entropy e_j for each criterion is computed to assess information diversity:

$$e_j = -k \sum_{i=1}^m r_{ij} \ln(r_{ij}), \quad k = \frac{1}{\ln(m)} \quad (8)$$

Weights w_j are derived based on the entropy values, highlighting the relative importance of criteria:

$$w_j = \frac{1 - e_j}{n - \sum_{j=1}^n e_j} \quad (9)$$

4.3.2 Technique for Order of Preference by Similarity to Ideal Solution

The TOPSIS[4] method is a multi-criteria approach to quantify the indicators. It evaluates a set of alternatives based on their distance from the most ideal solution and the most negative ideal solution. Let there be m alternatives from dataset and n criteria. The decision matrix is normalized to allow comparisons across criteria using:

$$P_{ij} = \frac{d_{ij}}{\sqrt{\sum_{i=1}^m d_{ij}^2}} \quad (10)$$

The weighted normalized decision matrix Weights w_j are applied to each criterion to reflect its importance, with $\sum_{j=1}^n w_j = 1$. Then, the positive (A^+) and negative (A^-) ideal solutions are defined as follows:

$$A^+ = \{\max_i v_{ij} | j \in J_{\text{benefit}}, \min_i P_{ij} | j \in J_{\text{cost}}\}, \quad (11)$$

$$A^- = \{\min_i v_{ij} | j \in J_{\text{benefit}}, \max_i P_{ij} | j \in J_{\text{cost}}\}, \quad (12)$$

where J_{benefit} includes all criteria where a higher value is considered better or more desirable and J_{cost} encompasses criteria where a lower value is preferred.

Given a alternative D , calculate its $L2$ distance from A^+ (D^+) and A^- (D^-):

$$D_i^+ = \sqrt{\sum_{j=1}^n (P_{ij} - A_j^+)^2}, \quad D_i^- = \sqrt{\sum_{j=1}^n (P_{ij} - A_j^-)^2} \quad (13)$$

The closeness coefficient CC_i determines the relative proximity to the ideal solution:

$$CC_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (14)$$

The combined Entropy-TOPSIS method offers a structured and objective framework for complex decision-making scenarios, effectively balancing the need for objective criteria weighting with comprehensive alternative evaluation. Hence, we build the base score function $F_{\text{base_score}}$ based on this method.

4.3.3 Indicator Establishment

We now introduce the indicators we used in this Entropy-TOPSIS method. To evaluate the actual performance of players in a given time, regardless of the initial advantages from severing rights, we summarized and processed five individual indicators, eliminating the serving effect, to evaluate the base score of performance as follows:

- x_{b1} : Quantized speed of serving.
- x_{b2} : Whether won this point.
- x_{b3} : The run distance per rally.
- x_{b4} : The count of the rally.
- x_{b5} : The count of the error serving.

These indicators are processed to regardless the initial advantages or disadvantages. Specifically, for data 'speed_mph' we eliminate the effect of serving, which is quantized as follow:

$$quantized_speed = \begin{cases} speed_mph - average_speed & \text{if server} \\ 0 & \text{else} \end{cases} \quad (15)$$

The quantized speed can precisely represent the performance of serving. A relative high speed serve contributes the performance score. Still, for not serving situation, this indicator won't damage the performance as well.

We then introduce the indicator 'distance_run_per_rally' is calculate with the run distance dividing the counts of the rally, and the indicator 'serve_errors_no' is a combination of the data 'serve_no' and 'double_fault'.

4.3.4 Bonus & Penalty Mechanism

As shown in Table 4, the given data consist of 3 win conditions and 3 loss conditions. One score match only satisfies one of these situation, causing these data largely remained zero. Thus, it requires large data amount to learn these conditions. Furthermore, these win or loss conditions highly implying the final result of current match. However, the performance evaluation model should take all considerations of the actual performance of players during the game, instead of only based on the final result.

Table 4: Bonus & Penalty Conditions.

Win or Loss Condition	Explanation	Score
Untouchable Winner	Indicates whether Player 1 hit an untouchable winning shot.	+5
Ace Serve	Indicates whether player hit an untouchable winning serve.	+5
Net Point Won	Indicates if player won the point while at the net.	+5
Unforced Error	Indicates if player made an unforced error.	-5
Net Approach	Indicates if player made it to the net during the point.	-5
Double Fault	Indicates if player missed both serves and lost the point.	-5

We take these win or loss conditions to consist an extra bonus and penalty score F_{bonus_score} adding behind F_{base_score} ranging from -5 – +5. The final performance score function as defined as follows:

$$F_{score}(x_b, x_c) = F_{base_score}(x_c) + F_{bonus_score}(x_c), \quad (16)$$

where x_b represents the indicators x_{b1} - x_{b5} and x_c are those winning or losing conditions. **The final results of $F_{score}(x_b, x_c)$ indicates the absolute performance of a player in a specific**

time. By comparing the results from two player we can evaluate how much better they are performing.

4.4 Results and Analysis for Model2

To precisely evaluate the difference between players' performance, as well as **how much better they are performing**, we propose a performance score evaluation model defined by equation 16, which consists of a base score function $F_{base_score}(x_1)$ solved with Entropy-TOPSIS method, and a bonus score function $F_{bonus_score}(x_2)$ considering extra conditions.

We solved this function using the data from the first match with the entropy method, which introduced in Section 4.3.1, and applies TOPSIS method to evaluate the closeness between the results and the ideal solution. The solved weights are listed in Table 5, where **we find the indicator 'rally_count' has more contribution than 'point_vector'**, which indicates that **our model has a balance consideration of performance based on various aspects, not only the victory**.

Table 5: Model weights proportion

Weights	Indicator Name	Value%
w_1	quantized_speed	5.495
w_2	point_vector	27.908
w_3	distance_run_per_rally	2.013
w_4	rally_count	38.029
w_5	serve_errors_no	3.044

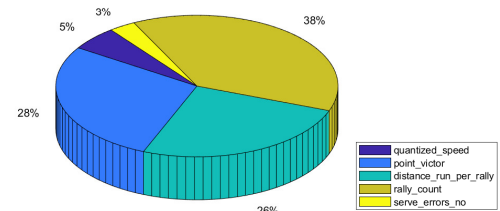


Figure 5: Proportion of weights

The final performance scores F_{score} were calculated based on the scoring and penalty items, combining the solved base score function F_{base_score} and the predefined bonus/penalty function F_{bonus_score} as follows:

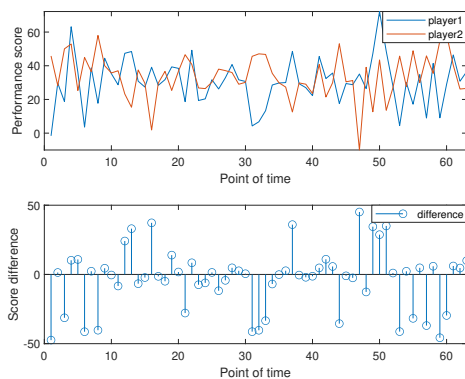


Figure 6: Evaluated performance score.

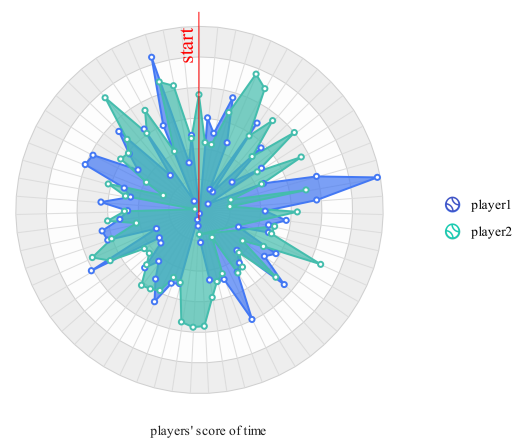


Figure 7: Radar chart of the performance score.

We visualized both results of the absolute performance score and its difference of two players during one set, in Figure 7. The intersection and difference of these performance score indicates who and how much better they are performing. The higher positive difference indicating a better performance of Player 1 compared to Player 2, leading to a better chance of winning. However, there are some cases that Player 1 lose the game with a better per-

formance, which might occur due to luck. Furthermore, both two players' performance score consist of a large trend, which is considered as the effect of 'momentum'.

5 Task2: Momentum Evaluation

Based on the evaluation of players' performance, we evaluate the 'momentum' during game flow by applying a time sequence model **LSTM** to predict the variations in the performance score through time with extra time sequence indicators, and smooth down the predicted results to represent the 'momentum' of players. By comparing the predicted 'momentum' from both players with the actual win or loss results, we **evaluate the effects of 'momentum' on the actual win or loss results**.

5.1 Momentum Prediction Model Based on LSTM

First, we introduced the LSTM time series model. Then, we established time-dependent features that affect the 'momentum', including the current cumulative score and the outcome of the previous set, etc. By constructing the objective function and loss function and incorporating them into the LSTM, we conducted predictive analysis.

5.1.1 Long-Short Term Memory

The LSTM[5] is a variant of RNN. There are four special structures in each repeated module, which interact specially. In Figure 8, each thin line transmits a whole vector, the yellow circle represents an operational operation, and the blue matrix represents the learned neural network layer. The core idea of LSTM model is "cell state". The "cell state" is similar to a conveyor belt, running directly on the whole chain with only a small amount of linear interaction and less information loss. In i_t, f_t, o_t, c_t, h_t and c_t respectively represent the input gate, forgetting gate, output gate, cell state, memory, and candidate state.

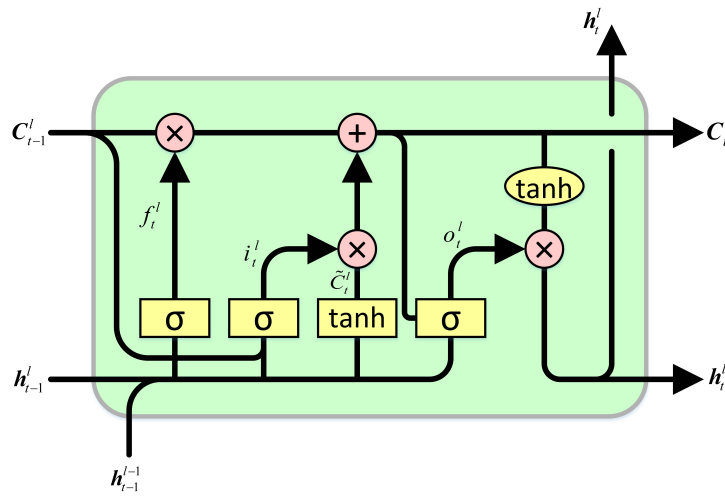


Figure 8: LSTM model visualization.

The LSTM works in following steps:

Step 1: Decide the discarded information in the cell. This operation is completed by the forgetting gate. First, read the current input x_t and previous neuron's output h_{t-1} , and the forgetting gate f_t decides the discarded information. The specific calculation formula is:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (17)$$

Step 2: Determine the new information stored in the cell state, in which sigmoid layer is used as the "input gate layer" to determine the updated value, and tanh layer creates a new candidate value vector $\tilde{C}_t$ to add to the state. The specific calculation formula is:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (18)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (19)$$

Step 3: Update the cell status, and update C_{t-1} to C_t . Multiply the old state with f_t , discard the information that needs to be discarded, and determine a new candidate value $i_t * \tilde{C}_t$. The specific calculation formula is :

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (20)$$

Step 4: Finally, the output gate decides what the next hidden state should be, which is based on the cell state but will be a filtered version.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (21)$$

$$h_t = o_t * \tanh(C_t) \quad (22)$$

5.1.2 Indicator Establishment

Adding extra indicators: Besides those indicators x_{b1} - x_{b5} applied in Model2 for performance evaluation, we add extra time sequence data including:

- x_{b6} : Current won score difference.

The indicator 'Current won score difference' considers the effects of scores won in the past and the current score difference between the two players. We sum the individual data points won_sets , won_games , and won_scores for both players in the current match as follows:

$$Total_Score = won_sets \times 20 + won_games \times 10 + won_scores, \quad (23)$$

here, we quantized the data 'NA' from won_scores as 50, and we times extra weights for won_sets and won_games to balance the value of each won data, as the quantized won_scores are ranging from 0-50. Then, we calculate the difference between two players to get the indicator 'Current won score difference'. The score difference between two players can largely affect the 'momentum' during the game flow.

- x_{b7} : Whether won the break point.

We also combined the data $p2_break_pt_won$ and $p2_break_pt_missed$ as x_7 'Whether won the break point', to indicate whether the player missed or caught the chance to win the set as a server. Specifically, we note positive 1 as not catching the chance, and in the note as negative 1 as he missed. The miss of such great chances can greatly harm the confidence and eventually affects the 'momentum'.

Predict function definition: As the effect of 'momentum' can be indicated from the variant of the performance score based on time. Our model directly predict the variations of the performance score, using those defined extra indicators and the performance score from current time. The predict function is defined as follow:

$$\hat{F}_{score} = F_{score} + LSTM(F_{score}, x_6, x_7), \quad (24)$$

where F_{score} indicates the performance score in current time, and $\hat{F}_{score}$ indicates the prediction of next point of time. The extra indicators x_6 and x_7 are also from the current time.

Loss function definition: We then define the loss function using RMSE to compare the difference between the prediction and true score as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (25)$$

where y_i is the true score difference and $\hat{y}_i$ is the predicted difference. By predicting the difference of each score, the model achieves a stable results, with more consideration of current score and focus on the variance of players performance.

5.2 Results and Analysis

Parameter settings and Model solving: We dived 90% of all data from the first match as train set, and other 10% as test set. We solved this model with Adam, a novel optimizer. In order to prevent gradient explosion, the gradient threshold is set to 1, and the initial learning rate is specified as 0.005. After half of the training, the learning rate is reduced by multiplying by a indicator of 0.2.

The effect of the 'momentum' can be indicated from the variant of the performance score through time. Hence, while Model2 giving the evaluation of the performance, we applies a time sequence model **LSTM to predict the variations of performance score** to evaluate the 'momentum'.

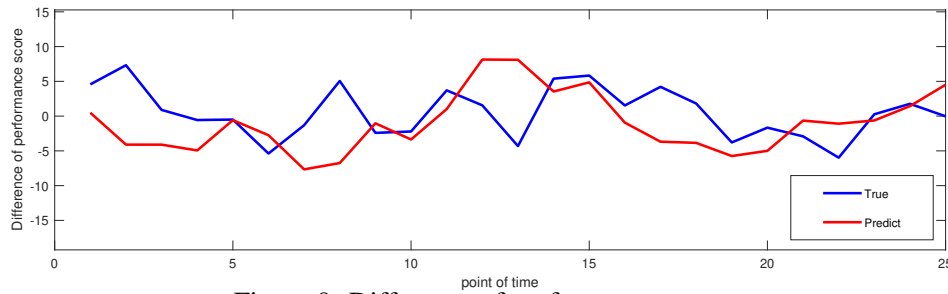


Figure 9: Difference of performance score.

As shown in Figure 9, where red line indicates the actual difference performance score per step and blue line is the predict one, also indicates as evaluated 'momentum'. All these data are generated from test data. The predicted momentum has the similar trends with the actual variation of performance score. **Precisely, our model achieved a 8.134 of the RMSE in the test set, which shows that our model is relatively reliable.**

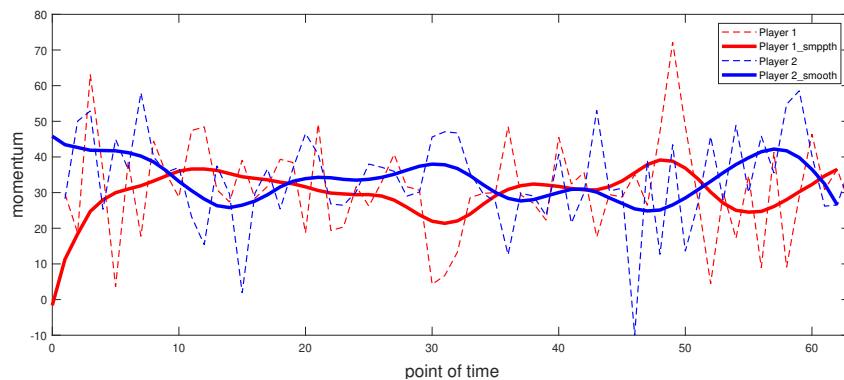


Figure 10: Visualization of the momentum for both two players.

To evaluate the flow of the 'momentum', we first set a equal initial performance score 30 for both two players, and adds predicted difference calculated by previews model individually

per point of time. Then, we smoothed down both two predicted performance score to show the trend of the 'momentum'. The results of both two players are shown in Figure 10, where the solid lines indicate the smoothed down trend of the 'momentum' and the dotted lines are the predicted performance score. **During the match, the 'momentum' of two players shows the flow of play changing from favoring one player to the other.**

We also visualized Player 1's 'momentum' in each set in Figure 11, where x axis represents each sets, y axis represents the time sequence and z axis is the 'momentum'. By comparing the 'momentum' from each sets we can clearly find a similar trend that: in a entire set, the 'momentum' of the Player 1 starts with a high point due to advantages of serving, then the 'momentum' variations with the game flow and getting high in the last few games contesting for key points.

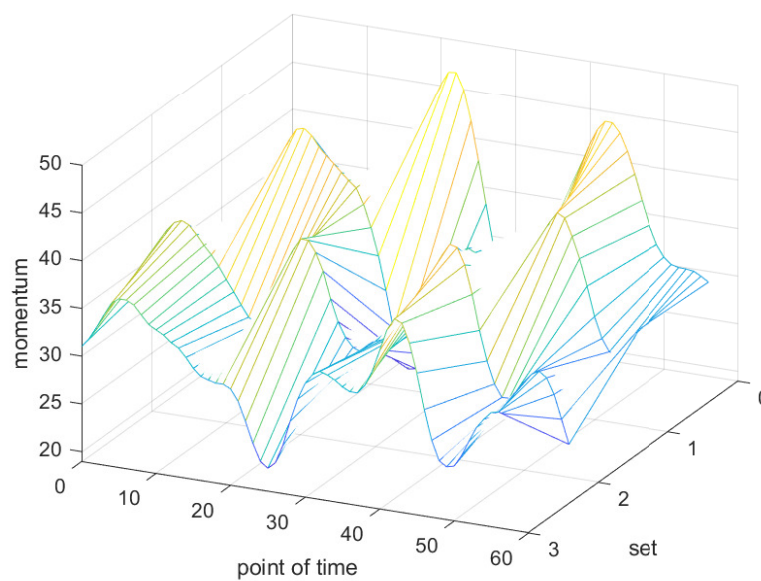


Figure 11: Visualization of the momentum for both two players.

To verify whether the 'momentum' actually affects the final win or loss, we compare the actual results from the first set of the first match with the random generated results. **We generate the random results** by adding individual random numbers per point of time to the initial performance score for both two players. Then we compare the random performance score to decide who performs better and won this score. By calculates through all times, we can generate a set of random scores based on random 'momentum'.

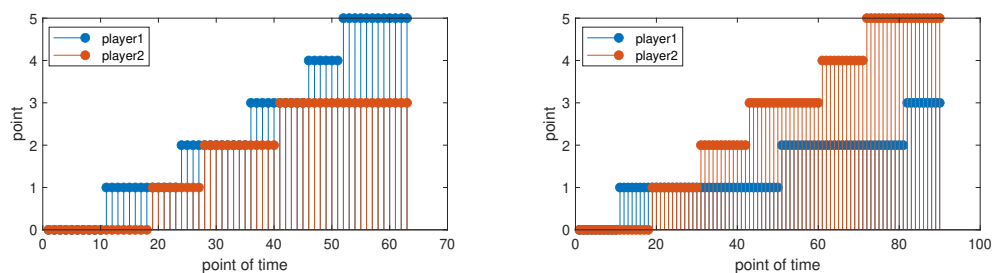


Figure 12: Compare of the Actual Wins with Random Wins.

As shown in Figure 12, the left side is the actual results of winning and the right side is the random generated one. The blue indicates Player 1's won games, the red indicates Player 2.

The extremely large difference between the Random and the Actual results shows that the 'momentum' acts a significant role in winnings, proving the wrong perspective of swings in play and runs of success by one player are random.

6 Task3: Momentum Swing Prediction

By assuming the crossing point of two players' momentum indicates the swing point, we applies the Random Forest Model to predict the future variations of the absolute difference between two players' momentum, and noting that the zero point of the this difference represents the swing point in a match. We find key indicators affecting the swing point through weights distribution of solved Random Forest Model and applies Spearman's rank to calculate the correlation coefficient. Then, we give our suggestions for players to maintain their advantages.

6.1 Random Forest

Random Forest[6] is a machine learning algorithm that operates by constructing multiple decision trees during the training phase and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. This method merges multiple randomized decision trees to form a single prediction by averaging the outcomes. It has proven to be exceptionally reliable, especially in scenarios where there are more variables than observations.

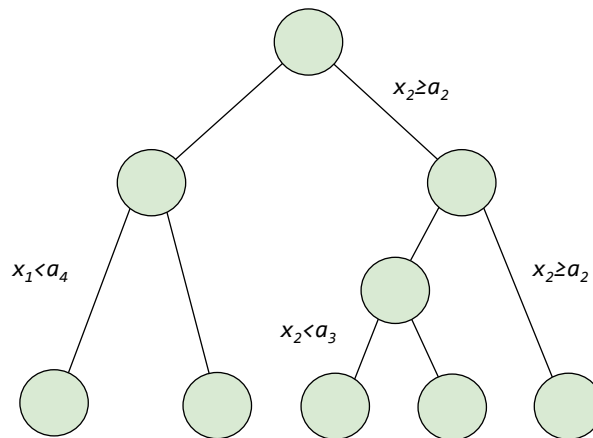


Figure 13: A graphical representation of the decision tree.

We applies the Random Forest to predict the swings in a match, by estimating the variations of the absolute difference between two players' momentum, assuming the crossing point of the momentum indicates the swing point. Specifically, To solve this problem, Random Forest model consist of the following steps:

Step1: It starts by creating N different bootstrap training samples from the original dataset, each of which is used to train a separate decision tree.

Step2: At each split in the training process, a random subset of m features is selected out of the total M features. The best split is found within this subset. This procedure adds diversity among the trees and reduces their correlation.

Step3: Each tree is grown to its fullest extent without pruning, resulting in complex, fully developed trees.

Step4: For *regression*, the prediction is the average of the outputs from all trees:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N y_i, \quad (26)$$

here, y_i represents the prediction of the i th tree, and N is the total number of trees in the forest.

6.2 Spearman's Rank Correlation Analysis

The Spearman's rank correlation coefficient[7], often denoted as Spearman's rho (ρ), is a non-parametric measure of rank correlation that assesses how well the relationship between two variables can be described using a monotonic function. It evaluates the statistical dependence between the rankings of two variables. Unlike Pearson's correlation coefficient, Spearman's rho does not assume that both datasets are normally distributed.

The full version of the calculation for Spearman's rank correlation coefficient can be considered in two parts. Firstly, when accounting for tied ranks, the formula is expressed in terms of the ranks of x and y :

$$\rho = \frac{\sum (R(x_i) - \bar{R}(x))(R(y_i) - \bar{R}(y))}{\sqrt{\sum (R(x_i) - \bar{R}(x))^2 \sum (R(y_i) - \bar{R}(y))^2}},$$

where:

- $R(x_i)$ and $R(y_i)$ are the ranks of x and y observations respectively,
- $\bar{R}(x)$ and $\bar{R}(y)$ represent the average ranks of x and y .

The more straightforward formula, used when there are no tied ranks, is shown below:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)},$$

where:

- $\sum d_i^2$ is the sum of the squares of the rank differences between paired observations,
- n is the number of pairs.

This formula can be simplified under certain conditions, such as when there are no tied ranks within the data. In the presence of tied ranks, the computation involves adjusting the ranks before calculating the differences, thereby complicating the process. The simplified form assumes all d_i (the differences in ranks between each pair of observations) are based on unique rank assignments without the need for tie adjustments.

6.3 Indicator Establishment

Here, we mainly applies predefined indicators from both players for Random Forest Model, We summarized several possible predefined indicators of two players that might affect this swing as follows:

- x_{c1} : Whether Player 1 is the server.
- x_{c2} : The count of the error serving of two players.
- x_{c3} : Quantized speed of the serve of two players.

- x_{c4} : The distance run per a rally of two players.
- x_{c5} : The count of the rally.

By applying these indicators for variation prediction Random Forest model, the solved weights can indicate the correlation between these indicators and the swings of 'momentum'. And for further evaluation, we applied Spearman's rank to calculate the correlation coefficient between these indicators and the swing point.

6.4 Results and Analysis

While previous model gave an evaluation of 'momentum', we predict the swings in match by predicting the future cross points of two players' 'momentum'. Specifically, we apply the Random Forest model to predict the variations of the two momentum's difference. The zero points of this difference indicate the crossing points of two momenta, also the swing points of the game. We divide 80% of all data from an entire match for training, and the other 20% for testing and visualizing the predicted results of the 'momentum' difference, comparing with the actual difference.

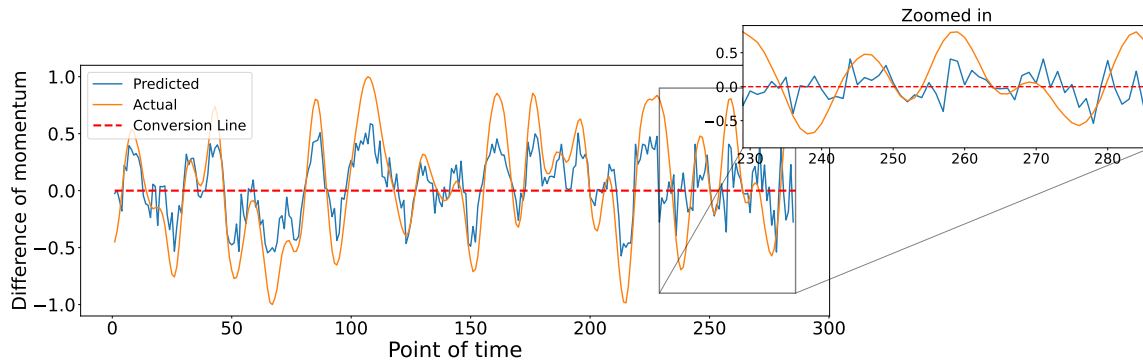


Figure 14: Predicted momentum difference, where zero points indicate swings in match.

As shown in Figure 14, the blue line is the predicted difference and the orange line is the actual difference. The zero points here indicate a crossing of these momentum, which also indicates a swing of the match flow. We zoomed in the prediction on the prediction for the test set, which closely follows the actual difference, and after smoothing, it accurately predicts a close enough swing point (zero point) indicating an excellent performance in predicting the future swings in match.

Then, we applied various error functions for validation, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and Coefficient of Determination (R^2).

Table 6: Model Results.

Dataset	MSE	RMSE	MAE	MAPE	R^2
Train Set	0.005	0.074	0.058	16.066	0.898
Test Set	0.035	0.187	0.149	40.485	0.068
Cross Validation	0.031	0.175	0.141	33.387	0.407

The validation results are shown in Table 6, where **our model ultimately achieved a low 0.035 of MSE, 0.187 of RMSE and 0.141 in MAE across the test data, showing its reliability on predicting the future trend of the momentum difference.** Still the results difference between the train set and the test set is narrowly become 0.03 on MSE, and 0.113 on RMSE, which also shows a high stability of our model.

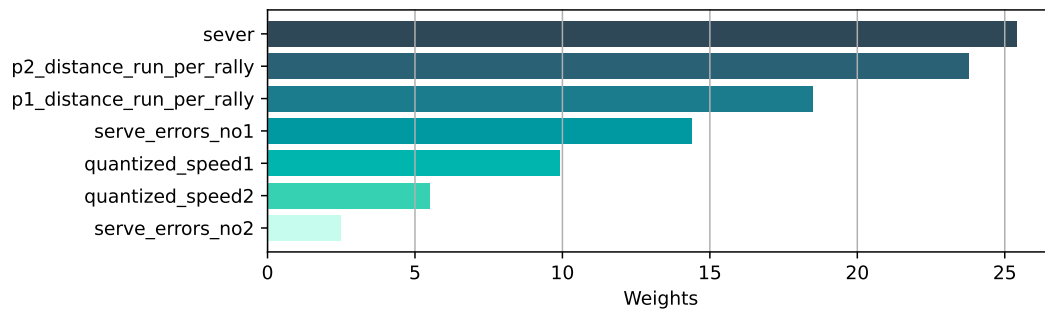


Figure 15: Correlation of indicators with the 'momentum' swings.

The solved weights indicate how each indicator correlates with 'momentum' swings. The proportion of each indicators are shown in Figure 15, where the alternate of serving can largely affects the game flow of winning. Still, the run distance per rally can also affect the swings of 'momentum'. The serve errors and serving speed has different effects on two players for helping them gain their momentum, which can be explained by: Each athlete has difference sensitivities to these indicators based on their own characteristics.

To more precisely evaluate the correlation of these indicators towards the swings in match, we further used Spearman's rank correlation to generate a heat map. Specifically, we evaluated the relationship of all indicators to the difference, calculated by dividing the momentum of Player 1 with Player 2. The positive value in the first row of the heat map indicates a positive effect on Player 1 to have a higher momentum.

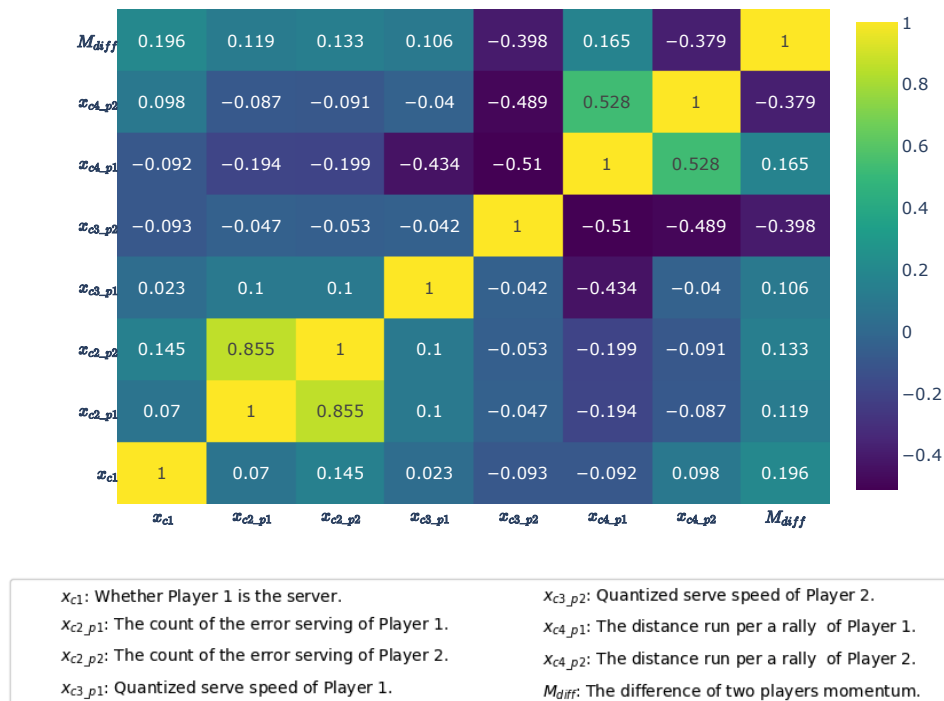


Figure 16: Correlation heat map of indicators.

As shown in Figure 16, the *longer run distance, slower serve speed and more serving mistakes* of Player 2 have positive effects on Player 1 to keep his advantage. Still, for Player 1, running less distance than the competitor, as well as having *faster serve and less mistakes* can largely help him to keep a high momentum. The most important indicator is *the right of serve*, showing a similar trend to previous results evaluations.

Suggestions: Based on the analysis above, we offer the following suggestions for athletes to maintain their advantage:

- The exchange of serving right has a significant impact on the shift of advantage. We suggest that athletes seize the advantage of good service rights while withstanding pressure when the opponent serves.
- The greater the distance covered during each stroke, the more energy is consumed, and the more passive the player becomes. We advise athletes to conserve energy appropriately during the match, control the pace of the game, and break free from being passive.
- For some athletes, mistakes made during the match may affect their mindset, thereby affecting their momentum. We recommend that athletes maintain a positive attitude, not be affected by mistakes, and keep up the momentum.

7 Task 4: Validation and Generalization Analysis

In this section, we test the generality of our swing point prediction model on various matches, including another Men's matches and a Women's matches. We also compare the difference between Men's matches and Women's matches of how those indicators affecting the swing in match.

7.1 Model Validation on Other Matches

We further conduct experiments on another tennis Men's matches '2023-wimbledon-1302', where 26 years old German player Alexander Zverev vs 27 years old Italian player Matteo Berrettini. In this match, the dominance between the players shifted very frequently following the shift of the serving right, consisting of 10 full win games.

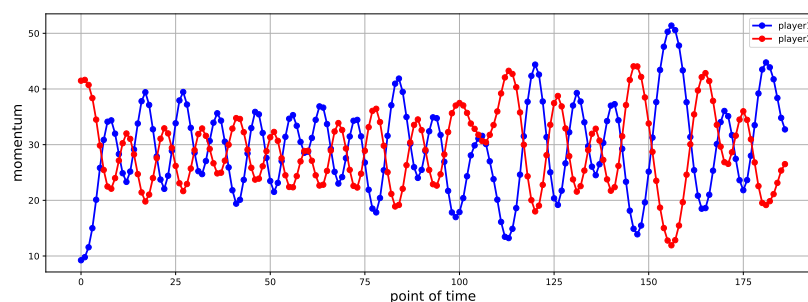


Figure 17: Momentum visualization for Men's matches.

To estimate the swing in players' 'momentum' during the game follow, we first apply the performance score model to evaluate the trend of both layers' 'momentum'. As visualized in Figure 17, where blue line indicates the momentum of Player 1 and red line indicates the one of Player 2, the crossing point of these two momentum appears frequently.

Then, we calculate the difference of these two 'momentum' and trains the swing point prediction model using the first 80% of data to predict the future swing points.

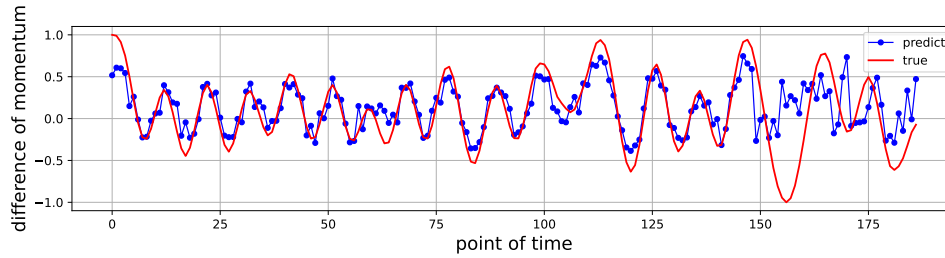


Figure 18: Swings in match: Momentum difference prediction for Men's matches.

The predicted results of the difference between two player's 'momentum' are shown in Figure 18, where blue line indicates the predicted difference and the red are the actual difference. While training with 160 time points, the model has closely fitted the true line in last 20 data. However, our model was hard to fit the data in 150-160, where the true line aggressively varies.

7.2 Generalization Analysis on Women's Matches

In this section we validate our model in a Women's matches to see its generality. We collect a similar data of Women's matches '2023-wimbledon-2503', where Elena Rybakina beaten by Ons Jabeur in quarter-finals, from the website '<https://www.atptour.com/en/scores>'. Women's matches has quite difference characters compared with Men's matches. For example, Woman's match consist of more rally count and more run distance compared to Men's matches, but having less chance of winning in Ace and slower serve speed.

We now calculates the momentum and estimate the swing points. Likewise, after smoothing down their calculated performance score as 'momentum', we applies our Swing Point Prediction Model to predict the difference of two players' momentum. We train the model with first 80% of data and validate in last 20%. The predicted results are visualized as follows.

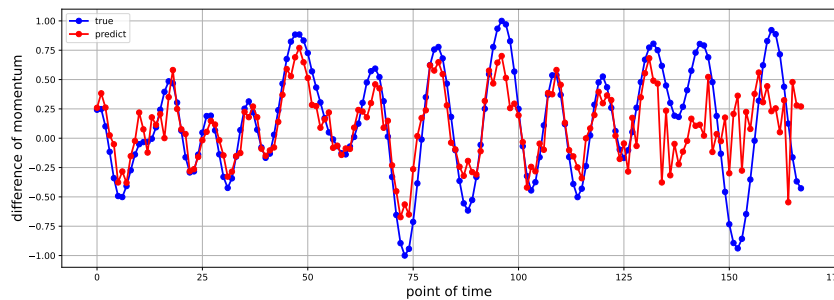


Figure 19: Swings in match: Momentum difference prediction for Women's matches.

As shown in Figure 19, where blue line indicates the predicted results and the red is the actual result, our model has an excellent fit in train data, but performs poorly while fitting the test data with aggressive variations. Especially for this match, that the vibration range is large and intense, remaining challenge for our model.

We also compare the difference between Men's matches and Women's matches of how those indicators affects the swing in match. The heat map of indicators and difference momentum are visualized as follow.

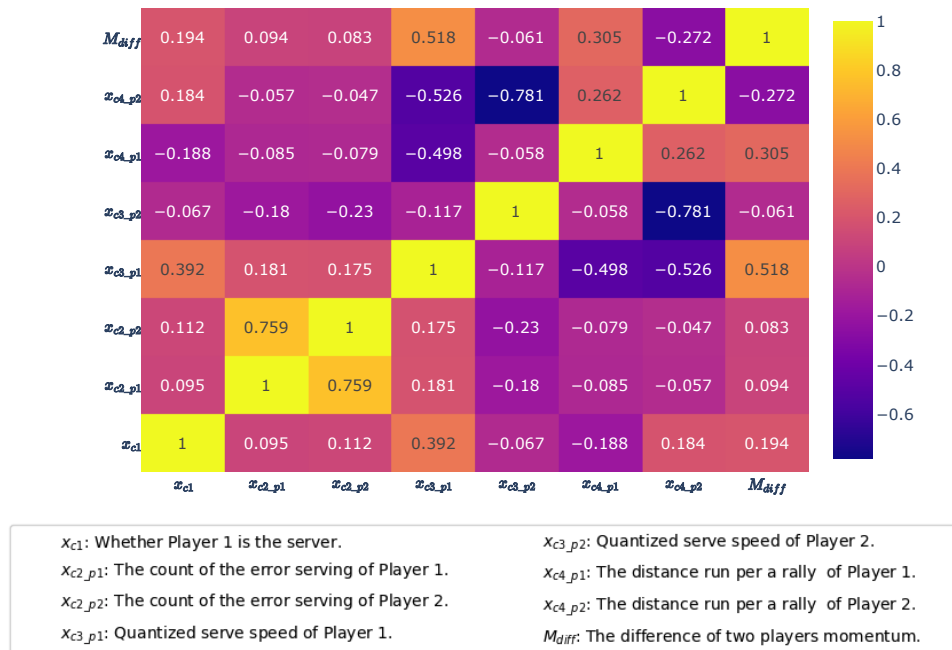


Figure 20: Correlation heat map of indicators in Women's matches.

As shown in Figure 20, the distribution of relevance for the indicator is very similar to that of Men's matches. Especially the most importance indicators including 'server' and run distance per rally. We then visualize the difference of these effects in Figure 21.

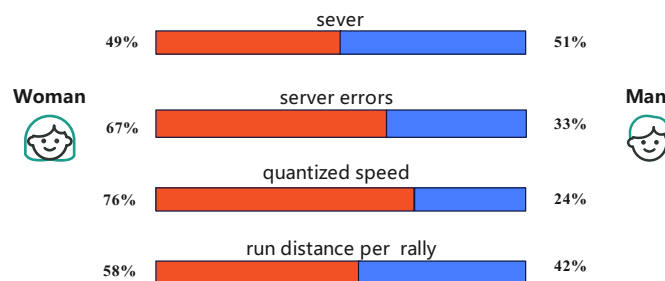


Figure 21: Compare of indicators effects in Men's and Women's matches.

Some indicators in Women's matches, like serve errors, has much larger negative effects on player's momentum compare to Men's matches, that having -0.092 in Men's matches and -0.188 in Women's matches. Moreover, indicators like serve speed also has larger effects on momentum in Women's matches than the Men's matches.

7.3 Model Improving

Based on above results in Men's and Women's match, we summarized the disadvantage of our model as hard to fit aggressive variations on momentum difference. To solve this problem, we proposed some extra factors that might helps our future models as follows:

- Service Record Year-to-Date:** A compilation of statistics that summarize a tennis player's serving performance over the current year, up to the present date. This record typically includes various metrics that quantify aspects of a player's serve, providing insights into their effectiveness, power, and overall contribution to their matches throughout the year.

- **Movement and Court Coverage** Using player tracking technology, it's possible to evaluate a player's speed, agility, and court coverage. Metrics can include average speed during play, distance covered per point or match, and efficiency of movement in terms of positioning and shot anticipation.
- **Return Efficiency** While basic return stats are common, return efficiency looks deeper into how well a player returns against first and second serves of varying speeds and spins, and how they convert defensive returns into offensive opportunities.
- **Rally Length Dominance** Analyzing performance based on rally length (short, medium, long) can highlight a player's strengths in different phases of play. For example, some players excel in quick points off serve, while others perform better in extended rallies.

8 MEMO

To: Tennis coaches

From: Chinese tennis lover

Date: 2024/2/5

Subject: Insights and Strategies from the 2023 Wimbledon Championships Analysis

Through the 2023 Wimbledon Championship final, where 20-year-old newcomer Carlos Alcaraz defeated seven-time Wimbledon champion Novak Djokovic, we observed an elusive momentum within players that significantly impacts the flow of matches. Mastering this momentum and formulating corresponding strategies during the game is crucial to the outcome. To this end, we have developed four models:

- A performance judgment model based on logistic regression, designed to predict the scoring opportunities for players at any given moment.
- A performance scoring model utilizing the entropy-weighted Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to concretize players' performances at specific time points.
- A momentum prediction model employing Long Short-Term Memory (LSTM) networks to forecast players' momentum throughout the match.
- A factor ranking model integrating Random Forest and Spearman's rank correlation analysis to identify and prioritize factors influencing critical junctures in the match.

We initiated our approach with a logistic regression-based performance judgment model to forecast scoring opportunities for players at specific moments, facilitating early strategic planning for subsequent gameplay. This model, refined through extensive regression training, has achieved a 73% accuracy rate in performance judgment. Subsequently, we employed an entropy-weighted TOPSIS-based performance scoring model for a granular analysis of player performances at given time points. This method enables a clearer differentiation between your player's performance and that of the opponent. We recommend considering these insights for optimizing game strategies and preparing more effectively for competitive engagements.

Subsequently, we employed an LSTM-based momentum prediction model to forecast the momentum of players throughout the match. This model has been validated to be accurate and reliable, enabling you to predict the overall trends of the match. It suggests adopting a strategy of pausing for adjustments when the opponent's momentum is on the rise, and seizing the initiative to press the advantage when your player's momentum is increasing.

Finally, we implemented a factor ranking model based on Random Forest to predict the transition points of player advantage changes, and apply Spearman's rank correlation analysis

to identify the main factors affecting these points. Accurate predictions of transition points through Random Forest enable you to foresee future changes in advantage and make strategic preparations in advance. Further examination using Random Forest and correlation analysis revealed that the server's role and the distance covered by players have the most significant impact on the match flow. Based on our analysis, we offer the following recommendations to maintain a competitive edge:

- The exchange of serving right has a significant impact on the shift of advantage. We suggest that athletes seize the advantage of good service rights while withstanding pressure when the opponent serves.
- The greater the distance covered during each stroke, the more energy is consumed, and the more passive the player becomes. We advise athletes to conserve energy appropriately during the match, control the pace of the game, and break free from being passive.
- For some athletes, mistakes made during the match may affect their mindset, thereby affecting their momentum. We recommend that athletes maintain a positive attitude, not be affected by mistakes, and keep up the momentum.

We hope these insights on momentum prove beneficial for the player's performance in matches. Thank you for considering our recommendations!

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